



3.23 COMPUTER STUDIES (451)

Computer Studies Paper

SECTION A (10 marks)

Answer all the questions in this section in the spaces provided.

Write each of the following acronyms in full as used in computing. (10 marks)

(a) CAD

(b) DVD

WORM

POS

A school keeps student records in a database. The data is coded before entry.

State three reasons why the coding is necessary. (3 marks)

Differentiate between Bcc and cc in an email. (2 marks)

State three risks posed by improper cabling in a computer laboratory. (3 marks)

List two career opportunities directly associated with computer networking. (2 marks)

A retailer uses a spreadsheet program to calculate profits. Figure 1 shows the spreadsheet.

	A	B	C	D	E	F
	Items	Cost Price	Selling Price	Profit per Item	Items sold	Total Profit
	Item 1	100	150	50	10	500
	Item 2	120	180	60	8	480
	Item 3	110	160	50	12	600
	Item 4	130	190	60	9	540

Figure 1

(a) Which row contains labels only? (1 mark)

Write the formula that has been entered in cell B4. (2 marks)

State three reasons why an organisation may opt to develop its own software in-house rather than buy off-the-shelf software. (3 marks)



□□ The topology below is formed by combining two types of topologies □

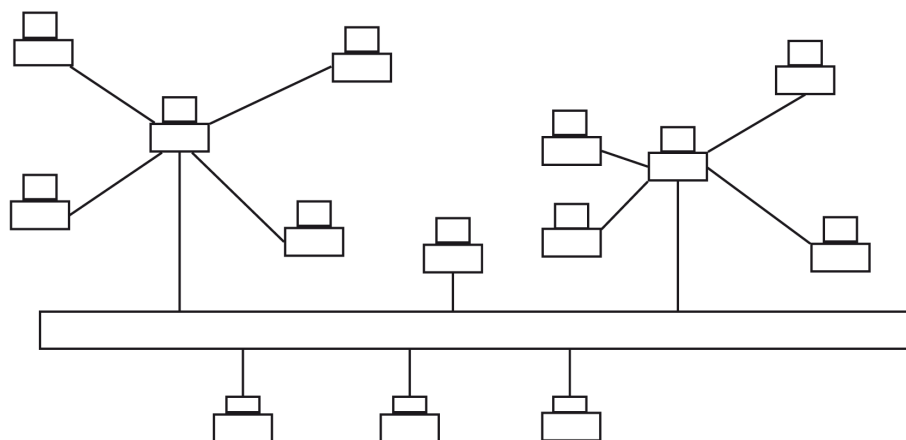


Figure 2

□ □
□ □ □ □ □ □ Figure □

□ a) Give one name for the combined topology in Figure 2 □ □ (1 mark)

□ b) Name the two topologies forming the combined topology in Figure 2 (1 marks)

□□ An organisation intends to replace an existing system by carrying out the process in □
□ stages (1)

□ a) Name this implementation strategy (1) □ □ (1 mark)

□ b) Give two reasons why the organisation is opting to use the implementation strategy in □
□ a) above (1) □ (1 marks)

□□ a) Explain the importance of disk partitioning (1) □ □ □ (1 marks)

(b) Differentiate between pull-down menu and pop-up menu as used in Graphical User
□ □ Interface GUI in operating systems (1) □ □ □ □ (1 marks)

□□ The 21st century has had many forms of ICT technologies improving the various means of □
communication. However, these changes have brought many challenges.
□ State three negative social impacts of these technologies (1) □ □ (1 marks)

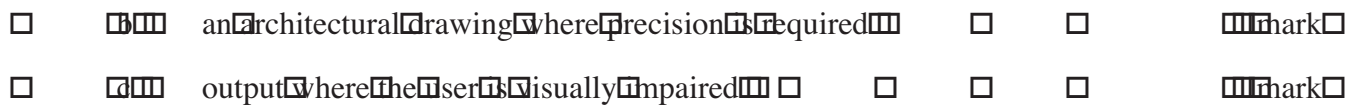
□□ Explain why a DTP application would be preferred to a word processing application when
□ designing a publication (1) □ □ □ □ □ □ □ □ (1 marks)

□□ A computer is idle but the hard disk might be linking indicating some activity □
□ State two possible causes of this (1) □ □ □ □ □ □ (1 marks)

□□ Describe **compatibility** as a factor to consider when purchasing a computer (1 marks)

□□ Identify the appropriate output device for the production of each of the following □

□ a) receipts where carbon copies are required (1) □ □ □ □ (1 mark)



Answer question **1** and any other **THREE** questions from this section in the spaces provided.

```

graph TD
    Start([START]) --> Read[/Read X/]
    Read --> Sum0[Sum = 0]
    Sum0 --> N0[N = 0]
    N0 --> LoopStart(( ))
    LoopStart --> Nplus[N = N + 1]
    Nplus --> Sumplus[Sum = Sum + N]
    Sumplus --> Decision{N = X}
    Decision -- No --> LoopStart
    Decision -- Yes --> Print[/Print Sum/]
    Print --> End([END])

```

☐	(i)	Determine the output from the flowchart if:										
☐	☐	☐		X		☐	☐	☐	☐	☐	☐	
☐	☐	☐		X		☐	☐	☐	☐	☐	☐	
☐	☐	☐	☐	☐	☐							
☐	☐		Write a Pseudocode for the flowchart in figure 3			☐	☐					
☐	☐		Modify the flowchart so that it can be used to get the sum of integers between									
☐	☐	☐		and		☐	☐	☐	☐	☐	☐	
☐	☐											
☐		List two programming language translators				☐	☐	☐				

